Reading project in Harmonic Analysis

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1 Introduction

Harmonic analysis, rooted in the classical study of Fourier series, has evolved into a central pillar of modern mathematics. At its heart lies the idea of understanding functions by decomposing them into basic oscillatory components—an approach that not only unveils deep structural properties of functions and spaces, but also unifies diverse fields ranging from number theory to representation theory.

We study Fourier Theory on the particularly nice class of functions, the Schwartz Space. Using this space makes the proofs much more elegant, and the results much more intuitive. We prove the Plancherel Theorem and the Poisson Summation Formula for the Fourier Transform on the Schwartz Space. We also generalize function spaces to the space of distributions, and extend the Fourier Transform to this space.

Our next goal is to generalize this to Locally Compact Abelian Groups, on which we show the beautiful Pontryagin Duality. Using Haar Integration we show the general Plancherel Theorem for Locally Compact Abelian Groups. This concludes the completeness of the Fourier Series.

Throughout our discussion, we use mostly the Riemann Integral, intentionally avoiding the Lebesgue Integral. Our intention is to explore as much of the subject as we can while assuming only minimal background. One only needs understanding of basic analysis, group theory and some topology.

Much of this material is based on Anton Deitmar's A First Course in Harmonic Analysis. Proofs and definitions have been adapted with modifications and simplifications where appropriate.

2 Hilbert Spaces

We always assume that the underlying field is $\mathbb{C}$ and by *space*, we always mean a $\mathbb{C}$-**inner product space**.

Definition 1 (Hilbert Space)**.** A complete *space* is called a Hilbert space.

Proposition 1. A finite-dimensional *space* is complete.

Proof. Let V be a *space* of dimension n. Then, $V \cong \mathbb{C}^n$, which is complete. □

2.1 Orthonormal Basis

To accommodate ∞-dimensional *spaces*, we extend the definition of a basis.

Definition 2 (Orthonormal Basis)**.** An orthonormal basis of a *space* V, is a countable set of orthonormal vectors which span a dense subset of V.

Proposition 2. A *space* is separable $\Leftrightarrow$ it has an ONB.

Proof. Let D be a countable dense subset of the *space* V. Applying Gram-Schmidt Orthogonalization, we get the required.

Conversely, let (e_j) be an ONB of V. The following set is dense in V.

$$D = \left\{ \sum_{i=1}^{n} a_i e_i \mid a_i \in \mathbb{Q}(i), n \in \mathbb{N} \right\}$$

□

2.2 l^2 spaces

We show that the *space* of all "square-summable" functions form a Hilbert space.

Definition 3 (l^2 space)**.** Let S be an arbitrary set. $l^2(S)$ is the set of functions $f : S \to \mathbb{C}$ satisfying

$$\sum_{s \in S} |f(s)|^2 < \infty$$

One may also state the above sum as (where F is a finite subset of S).

$$\sum_{s \in S} |f(s)|^2 = \sup_{F \subseteq S} \sum_{s \in F} |f(s)|^2$$

Note that if S is finite, $l^2(S)$ is the *space* of all functions from S to $\mathbb{C}$, which is finite-dimensional. By **Proposition 1**, $l^2(S)$ is a Hilbert space.

Lemma 1. For any set S and $f, g \in l^2(S)$, the following inner product makes $l^2(S)$ a Hilbert space.

$$\langle f, g \rangle = \sum_{s \in S} f(s)\overline{g(s)}$$

Note that, the existence of the inner product induces the "square-sum" norm. By triangle inequality, $\|f + g\| \leq \|f\| + \|g\|$; thus, $l^2(S)$ is a *space*.

Proof. **Claim 1**: The inner product converges (i.e, $l^2(S)$ is a *space*).

Since, $|f(s)\overline{g(s)}| = |f(s)||g(s)|$, it suffices to prove it for real-valued non-negative functions f and g. Let F be a finite subset of S and $f_F := f \restriction F$ and $g_F := g \restriction F$. By Cauchy-Schwarz inequality,

$$\sum_{s \in F} f(s)g(s) = |\langle f_F, g_F \rangle| \leq \|f_F\|\|g_F\| \leq \|f\|\|g\|$$

which implies,

$$|\langle f, g \rangle| = \sum_{s \in S} f(s)g(s) = \sup_{F \subset S} \sum_{s \in F} f(s)g(s) \leq \|f\|\|g\| < \infty$$

Claim 2: Completeness

Let (f_n) be Cauchy in $l^2(S)$. Then $\forall s \in S$, $f_n(s)$ is Cauchy in $\mathbb{C}$. Hence, f_n converges pointwise, to say f.

Fix $\varepsilon > 0$. Let $n, m > N \in \mathbb{N} \Rightarrow \|f_n - f_m\|^2 < \varepsilon$. For $n \geq N$,

$$\sum_{s \in F} |f_n(s) - f(s)|^2 = \lim_{j \to \infty} \sum_{s \in F} |f_n(s) - f_j(s)|^2 \leq \sup_{j \geq N} \|f_n - f_j\|^2 \leq \varepsilon$$

Therefore $f_n \rightrightarrows f$ and $l^2(S)$ is a Hilbert space. □

2.3 Completion

Let V be an ∞-dimensional, seperable *space* with ONB (e_j). Akin to the finite-dimensional case, we have $\forall v \in V$,

$$v = \sum_{j=1}^{\infty} c_j e_j$$

where $c_j = \langle v, e_j \rangle$, since we can approximate v arbitrarily well by finite linear combinations of e_j's, i.e, $\overline{span(e_j)} = V$.

Theorem 1. The map $v \mapsto (c_j)$ is an isometry from V to $l^2(\mathbb{N})$.

Proof. $\forall v \in V$, let $c_j = \langle v, e_j \rangle$

$$0 \leq \left\| v - \sum_{j=1}^{n} c_j e_j \right\|^2 = \left\langle v - \sum_{j=1}^{n} c_j e_j, v - \sum_{j=1}^{n} c_j e_j \right\rangle = \|v\|^2 - \sum_{j=1}^{n} |c_j|^2$$

Let T be the linear map $v \mapsto (c_j)$. Thus, $\|Tv\|^2 = \sum_{j=1}^{\infty} |c_j|^2 \leq \|v\|^2 < \infty$.

Taking square roots of the above inequality, we get $\|Tv\| \leq \|v\|$.

For every $v \in \mathit{span}(e_j)$, we have $\|Tv\| = \|v\|$. Since the span is dense in V, the equality holds for all $v \in V$. □

Theorem 2 (Completion)**.** For every seperable *space* V, there is a unique Hilbert space H, and an isometry $T : V \to H$ that maps V to a dense subset of H.

Proof. This is true for general metric spaces, and hence in particular for seperable *spaces*. **Theorem 1** explicitly describes the isometry T. □

Notation. If V is a *space*, then we denote it's completion by $\overline{V}$

Corollary 1 (Fundamental Theorem of Infinite Dimensional Vector Spaces)**.** If H is an ∞-dimensional, seperable Hilbert space, then $H \cong l^2(\mathbb{N})$.

In conclusion, we have split Hilbert spaces into **3 categories**. The finite-dimensional ($\mathbb{C}^n$), the ∞-dimensional and seperable ($l^2(\mathbb{N})$) and the ∞-dimensional and inseperable.

2.4 Bessel's Inequality

The following is an important inequality about the coefficients along an ONB.

Theorem 3 (Bessel's Inequality)**.** Let (e_k) be an orthonormal sequence in a Hilbert space H. For every $x \in H$, we have

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Proof. By direct computation, for all $n \in \mathbb{N}$, we have

$$\begin{aligned} 0 \leq \left\| x - \sum_{k=1}^{n} \langle x, e_k \rangle e_k \right\|^2 &= \|x\|^2 - 2 \sum_{k=1}^{n} \mathrm{Re} \langle x, \langle x, e_k \rangle e_k \rangle + \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \\ &= \|x\|^2 - 2 \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 + \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \\ &= \|x\|^2 - \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \end{aligned}$$

□

Corollary 2 (Parseval's Identity)**.** Let (e_k) be an ONB of a Hilbert space H. For every $x \in H$, we have

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 = \|x\|^2$$

Remark. The above identity is analogous to the Pythagorean Theorem, which too asserts that the "square-sum of components of a vector along an ONB", is equal to the "square of the vector".

2.5 Convolutions

Notation. $L^1_{bc}(\mathbb{R})$ denotes set of all bounded, continuous functions $f : \mathbb{R} \to \mathbb{C}$ satisfying

$$\|f\|_1 := \int_{-\infty}^{\infty} |f(x)|dx < \infty$$

Note that, $L^1_{bc}(\mathbb{R})$ is **not** a *space* with $\|\cdot\|_1$. We introduce convolutions with domain $L^1_{bc}(\mathbb{R})$ mainly as a useful tool.

Definition 4 (Convolution)**.** For $f, g \in L^1_{bc}(\mathbb{R})$, define the convolution as,

$$f * g(x) := \int_{-\infty}^{\infty} f(y)g(x-y)dy$$

We justify the existence of the above integral. Let $|g| \leq C \in \mathbb{R}$, Then

$$|f * g(x)| \leq \int_{-\infty}^{\infty} |f(y)g(x-y)|dy \leq C \int_{-\infty}^{\infty} |f(x)|dx = C\|f\|_1$$

Proposition 3. Let $f, g \in L^1_{bc}(\mathbb{R})$. Then $f * g \in L^1_{bc}(\mathbb{R})$.

Proof. **Claim 1**: $f * g$ is continuous. (at $x_0 \in \mathbb{R}$)

Fix $\varepsilon > 0$. Assume $|g| \leq C$. By integrability of $|f|$, $\exists T > |x_0|$ such that

$$\int_{|y|>T} |f(y)|dy < \frac{\varepsilon}{4C}$$

$$\Rightarrow \int_{|y|>T} |f(y)||g(x-y) - g(x_0-y)|dy < 2C \int_{|y|>T} |f(y)|dy < \frac{\varepsilon}{2}$$

Within $[-T, T]$, by uniform continuity of g, $\exists \delta > 0$ such that $|x - x_0| < \delta$ implies

$$\int_{-T}^{T} |f(y)||g(x-y) - g(x_0-y)|dy \leq \frac{\varepsilon}{2\|f\|_1} \int_{-T}^{T} |f(y)|dy \leq \frac{\varepsilon}{2}$$

Thus,

$$|f * g(x) - f * g(x_0)| \leq \int_{-\infty}^{\infty} |f(y)||g(x-y) - g(x_0 - y)|dy < \varepsilon$$

Claim 2: $\|f * g\|_1 < \infty$

$$\begin{aligned}
\|f * g\|_1 &= \int_{-\infty}^{\infty} |f * g(x)|dx \\
&= \int_{-\infty}^{\infty} \left| \int_{-\infty}^{\infty} f(y)g(x-y)dy \right| dx \\
&\leq \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(y)g(x-y)|dydx \\
&= \int_{-\infty}^{\infty} |f(y)|dy \int_{-\infty}^{\infty} |g(x)|dx = \|f\|_1 \|g\|_1 < \infty
\end{aligned}$$

□

Property. The following are immediate.

$$f * g = g * f \qquad f * (g * h) = (f * g) * h \qquad f * (g + h) = f * g + f * h$$

2.6 Schwartz Space

The Schwartz space is a *space* of "rapidly decreasing" smooth functions, i.e functions with derivatives decreasing faster than any reciprocal of x.

Notation. For a multi-index $m, n \geq 0$, let $\sigma_{m,n}(f) := \sup_{x \in \mathbb{R}} |x^m f^{(n)}(x)|$

Definition 5 (Schwartz Space)**.** It is defined as the *space*,

$$\mathcal{S} = \{f : \mathbb{R} \to \mathbb{C} \mid f \text{ is } C^\infty, \sigma_{m,n}(f) < \infty \text{ for all integers } n, m \geq 0\}$$

with the inner product,

$$\langle f, g \rangle := \int_{-\infty}^{\infty} f(x)\overline{g(x)}dx$$

A C^∞ function is said to have *moderate growth* if all of its derivatives are dominated by polynomials, i.e $\forall k \geq 0, \exists l \in \mathbb{N}$ such that

$$\sup_{x \in \mathbb{R}} \frac{|f^{(k)}(x)|}{1 + x^{2l}} < \infty$$

Lemma 2. If f has *moderate growth*, then $f\mathcal{S} \subset \mathcal{S}$. Moreover if $g_j \xrightarrow{\mathcal{S}} g$, then $fg_j \xrightarrow{\mathcal{S}} fg$.

Proof. The first claim is obvious. For the second claim suppose $g_j \xrightarrow{\mathcal{S}} g$.

$$\begin{aligned}\sigma_{m,n}(fg_j - fg) &= \sup_{x\in\mathbb{R}} \left|x^m (fg_j - fg)^{(n)}\right| \\ &\leq \sup_{x\in\mathbb{R}} \left|(1+x^{2M}) \sum_{k=0}^{n} \binom{n}{k} \sigma_{m,n-k}(g_j - g)\right| \\ &\leq \sup_{x\in\mathbb{R}} \left|\sum_{k=0}^{n} \binom{n}{k} (\sigma_{m,n-k}(g_j - g) + \sigma_{m+2M,n-k}(g_j - g))\right|\end{aligned}$$

Note that M can be chosen since f has *moderate growth*. Choose $J \in \mathbb{N}$ such that for $j \geq J$, $\sigma_{p,q}(g_j - g) < \frac{\varepsilon}{2^{n+1}}$. Thus, for $j \geq J$, we have $\sigma_{m,n}(fg_j - fg) < \varepsilon$ and $fg_j \xrightarrow{\mathcal{S}} fg$. □

Convergence in $\mathcal{S}$

A sequence of functions f_n is said to converge to f in S if every derivative of f_n converges uniformly to that of f, faster than any polynomial.

$$\forall p, q \geq 0 \quad \sup_{x\in\mathbb{R}} |x^p (f_n - f)^{(q)}(x)| \xrightarrow[n\to\infty]{} 0$$

This is much stronger than uniform convergence.

Notation. We will use $f_n \xrightarrow{\mathcal{S}} f$ to denote convergence in $\mathcal{S}$.

2.7 Compactly supported smooth functions

$f : \mathbb{R} \to \mathbb{C}$ is said to have *compact support* or be *compactly supported* if there exists a compact $K \subset \mathbb{R}$ for which, $f(x) = 0$ when $x \notin K$.

Example. The standard bump function, with support $[-1, 1]$.

$$\varphi(x) = \begin{cases} e^{-\frac{1}{1-x^2}} & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

Notation. Let $\mathcal{D}$ denote the *space* of all compactly supported smooth functions, with the inner product from $\mathcal{S}$.

Convergence in $\mathcal{D}$

A sequence of functions f_n is said to converge to f in $\mathcal{D}$ if every derivative of f_n converges uniformly to that of f, and the support eventually lies in a compact set.

$$\exists N \in \mathbb{N}, \exists K \subset \mathbb{R} \quad \text{supp}(f_n) \subset K \text{ for } n \geq N \text{ and } \forall k \in \mathbb{N}, f_n^{(k)} \rightrightarrows f^{(k)}$$

Remark. From the definition, it follows that $\mathcal{D} \subset \mathcal{S}$. Further, we state without proof that $\mathcal{S}$ is **not** complete under norm induced by its inner product. Thus, $\mathcal{D} \subset \mathcal{S} \subset \overline{\mathcal{S}}$.

We now have the necessary tools to talk about Fourier transforms.

3 Fourier Series

Our main objective in this section is to decompose periodic functions into linear combinations of their frequencies. Note that a continuous function on $[0,1)$ defines uniquely, a period 1 function on $\mathbb{R}$. Thus, we begin by analyzing functions on this domain.

3.1 The Unit Torus

Consider the quotient group $\mathbb{R} \setminus \mathbb{Z}$. The cosets can be represented by $[0,1)$, but this is **not** an isomorphism. However, for the unit torus, $\mathbb{T} = \{e^{2\pi ix} \mid x \in \mathbb{R}\}$,

Observation. $\mathbb{R} \setminus \mathbb{Z} \cong \mathbb{T}$ via the map $x \mapsto e^{2\pi ix}$

Notation. Let $C(\mathbb{T})$ denote the set of all continuous functions $f : \mathbb{T} \to \mathbb{C}$.

Note that, $C(\mathbb{T})$ is a *space* via the inner product,

$$\langle f, g \rangle := \int_0^1 f(e^{2\pi it})\overline{g(e^{2\pi it})}dt$$

For $k \in \mathbb{Z}$, define $e_k : \mathbb{T} \to \mathbb{C}$ as the map $z \mapsto z^k$. Thus, $e_k \in C(\mathbb{T})$

Notation. Let $\|\cdot\|_2$ denote the norm induced by the above inner product and let $\xrightarrow{2}$ denote convergence of functions in this norm.

Theorem 4. (e_k) is an ONB of $\overline{C(\mathbb{T})}$.

Proof. The set (e_k) is orthonormal since $\langle e_i, e_j \rangle = \delta_{ij}$. For $c_k \in \mathbb{C}$, define

$$H := \left\{ \sum_{k\in\mathbb{Z}} c_k e_k : \sum_{k\in\mathbb{Z}} |c_k|^2 < \infty \right\}$$

The elements of H converge in $\|\cdot\|_2$. By **Theorem 2**, $\sum_{k\in\mathbb{Z}} c_k e_k \mapsto (c_k)$ is an isomorphism from H to $l^2(\mathbb{Z})$. Thus, $H = \overline{C(\mathbb{T})}$ and $\text{span}(e_k)$ is dense in H. □

Corollary 3. $\overline{C(\mathbb{T})} \cong l^2(\mathbb{Z})$

3.2 Fourier Series

By **Theorem 1**, every function in $\overline{C(\mathbb{T})}$ can be written as an infinite linear combination of e_k's. We make the following definition.

Definition 6 (Fourier Series)**.** The Fourier Series of $f \in \overline{C(\mathbb{T})}$ is defined

$$\mathcal{F}(f) := \sum_{k\in\mathbb{Z}} c_k e_k \text{ where } c_k = \langle f, e_k \rangle$$

Notation. $\mathcal{F}_n(f)$ denotes the nth partial sum of the Fourier series.

Clearly, for $f \in \overline{C(\mathbb{T})}$, we have $\mathcal{F}_n(f) \xrightarrow{2} f$. However, this does **not** imply pointwise convergence. On the other hand, $F_n(f) \rightrightarrows f$, implies both $\|\cdot\|_2$ and pointwise convergence.

Question. Under what conditions do we get uniform convergence?

Bessel's Inequality

The Bessel's Inequality on Hilbert spaces, along with **Theorem 1** implies

Theorem 5 (Bessel's Inequality)**.** For $f \in \overline{C(\mathbb{T})}$, we have

$$\sum_{k\in\mathbb{Z}} |c_k|^2 \leq \|f\|_2^2$$

3.3 Uniform Convergence

We show that **continuity** and **piecewise** C^1 guarantees uniform convergence.

Lemma 3. $f \in \overline{C(\mathbb{T})}$ be continuous and let (c_k) satisfy

$$\sum_{k\in\mathbb{Z}} |c_k| < \infty$$

Then $F_n(f) \rightrightarrows f$.

Proof. By *Weierstrass M-Test*, the series $\sum_{k\in\mathbb{Z}} c_k e_k$ converges uniformly to a continuous function, say g. Since $\mathcal{F}_n(f) \xrightarrow{2} f$, it follows

$$\|f - g\|_2 = 0$$

Since f, g are continuous, $f = g = \sum_{k\in\mathbb{Z}} c_k e_k$ □

Theorem 6. $f \in \overline{C(\mathbb{T})}$ be continuous, piecewise C^1. Then $F_n(f) \rightrightarrows f$.

Proof. Let c_k' be the Fourier coefficients of f'. Then

$$\sum_{k\in\mathbb{Z}} |c_k'|^2 \leq \|f'\|_2^2 < \infty$$

Using integration by parts, $c_k = \frac{1}{2\pi i k} c_k'$ and the GM $\leq$ AM inequality, we have

$$|c_k| \leq \frac{1}{2}\left(\frac{1}{4\pi^2 k^2} + |c_k'|^2\right) \Rightarrow \sum_{k\in\mathbb{Z}} |c_k| < \infty$$

By **Lemma 3**, we get $F_n(f) \rightrightarrows f$. □

4 Fourier Transform

We shall use weaker versions of *Lebesgue's dominated convergence theorem* and *Beppo-Levi's Lemma*, which we state without proof.

Notation. $f_n \xrightarrow{\text{l.u.c}} f$ denotes local uniform convergence, i.e, $\forall x \in \mathbb{R}$ there is a neighbourhood of x on which $f_n \rightrightarrows f$.

Definition 7 (Dominance)**.** A function g is said to dominate a sequence of functions $\{f_n\}$ if $\forall x \in \mathbb{R}, \forall n \in \mathbb{N}$, we have $|f_n(x)| \leq |g(x)|$.

Theorem 7 (Dominated Convergence)**.** Let f_n be a sequence of continous functions dominated by g such that $\int_{-\infty}^{\infty} g(x)dx < \infty$ and $f_n \xrightarrow{\text{l.u.c}} f$. Then

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f_n(x)dx = \int_{-\infty}^{\infty} f(x)dx$$

Theorem 8 (Monotone Convergence)**.** Let f_n be a sequence of contiuous, non-negative functions and $f_n \xrightarrow{\text{l.u.c}} f$ monotonically from below. Then

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f_n(x)dx = \int_{-\infty}^{\infty} f(x)dx$$

4.1 Definition

The notion of the Fourier transform is to break a function into its component "frequencies".

Definition 8 (Fourier Transform)**.** For $f \in \mathcal{S}$, we define its Fourier transform as

$$\hat{f}(y) := \int_{-\infty}^{\infty} f(x)e^{-2\pi ixy}dx$$

Property. The Fourier transform is bounded.

Proof. Let $f \in \mathcal{S}$ and assume $(1+x^2)f(x) \leq C$. Then

$$|\hat{f}(y)| \leq \int_{-\infty}^{\infty} |f(x)|dx \leq C\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx < \infty$$

□

Property. The following are immediate. Let $f \in \mathcal{S}$ and $a \in \mathbb{R}$,

(a) If $g(x) = f(x)e^{2\pi iax}$, then $\hat{g}(y) = \hat{f}(y-a)$.

(b) If $g(x) = f(x-a)$, then $\hat{g}(y) = \hat{f}(y)e^{-2\pi iay}$.

(c) If $g(x) = f(\frac{x}{\lambda})$, for $\lambda > 0$, then $\hat{g}(y) = \lambda \hat{f}(\lambda y)$

Property. The Fourier transform takes convolutions to pointwise product.

Proof. Let $h = f * g$. Then, by direct computation

$$\begin{aligned}\hat{h}(y) &= \int_{-\infty}^{\infty} h(x)e^{-2\pi ixy}dx \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x-z)g(z)dz\ e^{-2\pi ixy}dx \\ &= \int_{-\infty}^{\infty} f(x)e^{-2\pi ixy}dx \int_{-\infty}^{\infty} g(z)e^{-2\pi izy}dz = \hat{f}(y)\hat{g}(y)\end{aligned}$$

□

Property. The Fourier transform takes multiplication by $-2\pi ix$ to derivative.

Proof. By direct computation,

$$\hat{f}'(z) = \lim_{y\to z}\frac{\hat{f}(y)-\hat{f}(z)}{y-z} = \lim_{y\to z}\int_{-\infty}^{\infty} f(x)e^{-2\pi ixz}\frac{e^{-2\pi ix(y-z)}-1}{y-z}dx$$

The sequence, $\varphi_n(x) = \frac{e^{\frac{-2\pi ix}{n}}-1}{\frac{1}{n}} \xrightarrow{\text{l.u.c}} -2\pi ix$ and $|\varphi_n(x)| \le 2\pi|x|$. The claim follows by dominated convergence. □

Property. The Fourier transform takes derivative to multiplication by $2\pi iy$.

Proof. Using integration by parts,

$$\begin{aligned}\widehat{f'}(y) &= \int_{-\infty}^{\infty} f'(x)e^{-2\pi ixy}dx \\ &= e^{-2\pi ixy}\int_{-\infty}^{\infty} f'(x)dx + 2\pi iy\int_{-\infty}^{\infty} e^{-2\pi ixy}f(x)dx\end{aligned}$$

Since for $f \in \mathcal{S}$, we have $f(x) \longrightarrow 0$ as $|x| \longrightarrow \infty$, we get the required. □

Proposition 4. $\hat{\mathcal{S}} \subset \mathcal{S}$

Proof. Let $f \in \mathcal{S}$. By the above two properties we have,

$$((-2\pi ix)^n f)\hat{} = \hat{f}^{(n)} \quad \text{and} \quad \widehat{f^{(n)}}(y) = (2\pi iy)^n \hat{f}(y)$$

Thus, for $m, n \ge 0$, the function $y^m \hat{f}^{(n)}(y)$ is the Fourier transform of a function in $\mathcal{S}$, and hence bounded. □

Corollary 4 (Riemann-Lebesgue Lemma)**.** For $f \in \mathcal{S}$, $\lim_{|x|\to\infty} \hat{f}(x) = 0$.

4.2 Inversion Formula

The inversion formula shows establishes an inherent reversibility of the Fourier transform. It is the first step towards showing that the Fourier transform is a unitary map on $\overline{\mathcal{S}}$.

Define an auxiliary function, $h_\lambda(x) = \int_{-\infty}^{\infty} e^{-\lambda|t|} e^{2\pi i t x} dt$, for $\lambda > 0$.

Observation. $0 < e^{-\lambda|t|} \leq 1$ and $e^{-\lambda|t|} \xrightarrow[\lambda \to 0]{\text{l.u.c}} 1$.

Lemma 4. We have

$$h_\lambda(x) = \frac{2\lambda}{4\pi^2 x^2 + \lambda^2} \text{ and } \int_{-\infty}^{\infty} h_\lambda(x) dx = 1$$

and in particular, $h_\lambda(x) = \frac{1}{\lambda} h_1\left(\frac{x}{\lambda}\right)$ for every $\lambda > 0$.

Proof. The first claim follows by direct computation, and

$$\int_{-\infty}^{\infty} h_\lambda(x) dx = \frac{2}{\lambda} \int_{-\infty}^{\infty} \frac{1}{1 + \left(\frac{2\pi x}{\lambda}\right)^2} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1}{1 + x^2} dx = 1$$

□

Lemma 5. For every $f \in \mathcal{S}$ and $\lambda > 0$,

$$f * h_\lambda(x) = \int_{-\infty}^{\infty} e^{-\lambda|t|} \hat{f}(t) e^{2\pi i x t} dt$$

Proof. By direct computation,

$$\begin{aligned} f * h_\lambda(x) &= \int_{-\infty}^{\infty} f(y) h_\lambda(x - y) dy \\ &= \int_{-\infty}^{\infty} f(y) \int_{-\infty}^{\infty} e^{-\lambda|t|} e^{2\pi i t(x-y)} dt dy \\ &= \int_{-\infty}^{\infty} e^{-\lambda|t|} e^{2\pi i x t} \int_{-\infty}^{\infty} f(y) e^{-2\pi i t y} dy dt \end{aligned}$$

□

Lemma 6. For every $f \in \mathcal{S}$ and $x \in \mathbb{R}$,

$$\lim_{\lambda \to 0} f * h_\lambda(x) = f(x)$$

Proof. Using **Lemma 4**,

$$\begin{aligned} f * h_\lambda(x) - f(x) &= \int_{-\infty}^{\infty} f(y)h_\lambda(x-y)dy - \int_{-\infty}^{\infty} f(x)h_\lambda(y)dy \\ &= \int_{-\infty}^{\infty} (f(x-y) - f(x))h_\lambda(y)dy \\ &= \int_{-\infty}^{\infty} (f(x-y) - f(x))\frac{1}{\lambda}h_1\left(\frac{y}{\lambda}\right)dy \\ &= \int_{-\infty}^{\infty} (f(x-\lambda y) - f(x))h_1(y)dy \end{aligned}$$

Let $|f(x)| \leq C \in \mathbb{R}$. Then, $f(x-\lambda y) \xrightarrow[\lambda\to 0]{\text{l.u.c}} f(x)$ and the integrand is dominated by $2Ch_1(y)$. The claim follows by the dominated convergence theorem. □

Theorem 9 (Inverse Transform)**.** For $f \in \mathcal{S}$, we have $\hat{\hat{f}} = f(-x)$.

Proof. By **Lemma 5**, for $\lambda > 0$

$$f * h_\lambda(x) = \int_{-\infty}^{\infty} e^{-\lambda|t|}\hat{f}(t)e^{2\pi ixt}dt$$

By taking limit as $\lambda \to 0$, and using the fact that RHS is dominated by $|\hat{f}(t)|$, we get the required. □

Notation. The above theorem, allows us to define the *inverse transform* of $f \in \mathcal{S}$, which we denote by $\check{f}$.

$$\check{f}(x) := \int_{-\infty}^{\infty} f(y)e^{2\pi ixy}dy$$

Remark. With the *inverse transform* defined, we just need injectivity to ensure it is bijective. In fact, we can show the much stronger result, unitarity.

4.3 Plancherel's Theorem

In this section, we extend the *Fourier transform* to a unitary map on $\overline{\mathcal{S}}$.

Theorem 10 (Plancherel)**.** For every $f \in \overline{\mathcal{S}}$, we have

$$\|f\|_2 = \|\hat{f}\|_2$$

Proof. Let $\tilde{f}(x) = \overline{f(-x)}$ and $g = \tilde{f} * f$. Then

$$g(x) = \int_{-\infty}^{\infty} \overline{f(y-x)}f(y)dy$$

We have, $\hat{g} = \widehat{\overline{\hat{f}}} \cdot \hat{f} = \overline{\hat{f}} \cdot \hat{f} = |\hat{f}|^2$; via $\widehat{\overline{\hat{f}}}(x) = \hat{\overline{f}}(-x) = \overline{\check{f}}(-x) = \overline{\hat{f}}(x)$. Thus,

$$\begin{aligned}\|f\|_2^2 = g(0) &= \lim_{\lambda \to 0} g * h_\lambda(0) \\ &= \lim_{\lambda \to 0} \int_{-\infty}^{\infty} e^{-\lambda|t|} \hat{g}(t) dt \\ &= \lim_{\lambda \to 0} \int_{-\infty}^{\infty} e^{-\lambda|t|} |\hat{f}(t)|^2 dt = \|\hat{f}\|_2^2\end{aligned}$$

by the monotone convergence theorem. □

Thus, the Fourier transform is an isometry, and hence injective. By the denseness of $\mathcal{S}$ in $\overline{\mathcal{S}}$, $f \mapsto \hat{f}$ extends to a unitary map on $\overline{\mathcal{S}}$.

Corollary 5. The Fourier transform is a bijection on $\mathcal{S}$.

4.4 Poisson Summation Formula

The Poisson Summation Formula, connects the Fourier coefficients of a function to that of its transform.

Theorem 11 (Poisson Summation)**.** Let $f \in \mathcal{S}$. Then

$$\sum_{k \in \mathbb{Z}} f(k) = \sum_{k \in \mathbb{Z}} \hat{f}(k)$$

Proof. Define $F(x) := \sum_{k \in \mathbb{Z}} f(x+k)$. Note that F is 1-periodic. Let c_k be the Fourier coefficients of F.

$$\begin{aligned}c_k &= \int_0^1 \sum_{n \in \mathbb{Z}} f(x+n) e^{-2\pi i k x} dx = \sum_{n \in \mathbb{Z}} \int_0^1 f(x+n) e^{-2\pi i k x} dx \\ &= \sum_{n \in \mathbb{Z}} \int_n^{n+1} f(u) e^{-2\pi i k (u-n)} du = \left(\sum_{n \in \mathbb{Z}} e^{2\pi i k n} \right) \int_n^{n+1} f(u) e^{-2\pi i k u} du \\ &= \int_{-\infty}^{\infty} f(u) e^{-2\pi i k u} du = \hat{f}(k)\end{aligned}$$

Thus, $F(0) = \sum_{k \in \mathbb{Z}} f(k) = \sum_{k \in \mathbb{Z}} \hat{f}(k)$. □

The main conclusion is that the **Fourier transform is a unitary map on $\mathcal{S}$.** Later we shall try to extend the Plancherel Theorem to more general spaces.

5 Distributions

Distributions allow is to define a notion of a "generalized function" which acts on certain "test functions". For us, the "test functions" will be $\mathcal{D}$.

Definition 9 (Distribution)**.** A distribution is a continuous linear functional $T : \mathcal{D} \to \mathbb{C}$, i.e,

$$\lim_{n\to\infty} T(g_n) = T(g) \text{ whenever } g_n \xrightarrow{\mathcal{D}} g$$

Example.

- Dirac delta: $\delta(f) := f(0)$
- Integral: $I(f) := \int_{-\infty}^{\infty} f(x)dx$
- Let $\phi : \mathbb{R} \to \mathbb{R}$ be locally Riemann integrable. Then $I_\phi(f) := \int_{-\infty}^{\infty} f(x)\phi(x)dx$.

Notation. Let $\mathcal{D}'$ denote the space of all distributions.

We motivate a general definition of a derivative. Let $\phi \in C^1(\mathbb{R})$. Using integration-by-parts,

$$I_{\phi'}(g) = \int_{-\infty}^{\infty} \phi'(x)g(x)dx = -\int_{-\infty}^{\infty} \phi(x)g'(x)dx = -I_\phi(g')$$

Definition 10 (Derivative of a distribution)**.** Let $T \in \mathcal{D}'$. We define it's derivative $T' \in \mathcal{D}'$ as

$$T'(g) := -T(g')$$

5.1 Tempered Distributions

The tempered distributions are a "well-behaved" class of distributions to work with.

Definition 11 (Tempered Distribution)**.** A tempered distribution is a continuous linear functional $T : \mathcal{S} \to \mathbb{C}$, i.e

$$\lim_{n\to\infty} T(g_n) = T(g) \text{ whenever } g_n \xrightarrow{\mathcal{S}} g$$

Notation. Let $\mathcal{S}'$ denote the *space* of all tempered distributions.

Proposition 5. The restriction $T \mapsto T \restriction_{\mathcal{D}}$ is injective. Thus $\mathcal{S}'$ can be considered a subspace of $\mathcal{D}'$.

Proof. Let $T \in \mathcal{S}'$ with $T \restriction_{\mathcal{D}} = 0$. We want to show that $T(f) = 0$ for all $f \in \mathcal{S}$.

Let $\eta : \mathbb{R} \to [0,1]$ be smooth, with

$$\eta(x) = \begin{cases} 0 & x \leq 0 \\ 1 & x \geq 1 \end{cases}$$

For $n \in \mathbb{N}$, set

$$\chi_n(x) := \eta(n+x)\eta(n-x)$$

Then χ_n has compact support and $\chi_n(x) = 1$ for $|x| \leq n-1$. Set $f_n(x) = \chi_n(x)f(x)$. Then each $f_n \in \mathcal{D}$ and $f_n \xrightarrow{\mathcal{S}} f$. Thus, $T(f) = \lim_{n\to\infty} T(f_n) = 0$. □

We show that for locally integrable functions ϕ, I_ϕ can generally be extended to a tempered distribution.

Lemma 7. Let ϕ be locally integrable and assume $k \in \mathbb{N}$ is such that

$$\int_{-\infty}^{\infty} |\phi(x)| \frac{1}{1+x^{2k}} dx < \infty$$

Then $I_\phi(f) = \int_{-\infty}^{\infty} \phi(x)f(x)dx$ converges for every $f \in \mathcal{S}$ and defines a tempered distribution.

Proof. The convergence of the integral is clear. Let $C = \int_{-\infty}^{\infty} |\phi(x)| \frac{1}{1+x^{2k}} > 0$ and suppose $f_k \xrightarrow{\mathcal{S}} f$. Let $\varepsilon > 0$. Then $\exists N \in \mathbb{N}$, such that for $n \geq N$, we have $\sup_{x\in\mathbb{R}} |f_n(x) - f(x)| < \frac{\varepsilon}{C(1+x^{2k})}$. The lemma follows. □

Proposition 6. The map $\phi \mapsto I_\phi$ from $\mathcal{S}$ to $\mathcal{S}'$ is injective, and extends to a natural embedding $\overline{\mathcal{S}} \hookrightarrow \mathcal{S}'$

Proof. Let $\phi \in \mathcal{S}$ with $I_\phi = 0$. For $f \in \mathcal{S}$, we have $0 = I_\phi(\overline{f}) = \langle \phi, f \rangle$. Since $\mathcal{S}$ is dense in $\overline{\mathcal{S}}$, we have $\phi = 0$, which establishes injectivity.

To see that it extends, let $\phi_n \longrightarrow \phi$ in $\mathcal{S}$ be Cauchy. Then, $I_{\phi_n}(f) = \langle \phi_n, \overline{f} \rangle$ and for $m, n \in \mathbb{N}$, we have

$$|I_{\phi_n}(f) - I_{\phi_m}(f)| = |\langle \phi_n - \phi_m, \overline{f} \rangle| \leq \|\phi_n - \phi_m\|_2 \|f\|_2$$

Thus, $I_{\phi_n}(f)$ is Cauchy in $\mathbb{C}$. Let $I_{\phi_n}(f) \longrightarrow I_\phi(f)$. We show that I_ϕ is continuous.

To this end, let $f_j \longrightarrow f$ in $\mathcal{S}$. Then $|I_\phi(f_j) - I_\phi(f)| \leq \|\phi\|_2 \|f_j - f\|_2$. Since convergence in $\mathcal{S}$ implies convergence in $\|\cdot\|_2$, we are done. □

5.2 Fourier Transform of Distributions

We motivate the definition for Fourier transform of tempered distributions. Let $\tilde{f}(x) = f(-x)$. By the inversion theorem, $\hat{\hat{f}} = \tilde{f}$. For $\phi, f \in \mathcal{S}$, we have

$$I_{\hat{\phi}}(f) = \left\langle \hat{\phi}, \overline{f} \right\rangle = \left\langle \hat{\hat{\phi}}, \hat{\overline{f}} \right\rangle = \left\langle \tilde{\phi}, \hat{\overline{f}} \right\rangle = \left\langle \phi, \tilde{\hat{f}} \right\rangle$$

Moreover,

$$\hat{\overline{f}}(x) = \int_{-\infty}^{\infty} \overline{f(y)} e^{-2\pi i x y} dy = \overline{\int_{-\infty}^{\infty} f(y) e^{2\pi i x y} dy} = \overline{\hat{f}(-x)}$$

Thus, $\hat{\hat{\overline{f}}} = \overline{\hat{f}}$, and so, $I_{\hat{\phi}}(f) = I_{\phi}(\hat{f})$.

Definition 12. Let $T \in \mathcal{S}'$. We define the Fourier transform of T as

$$\hat{T}(f) := T(\hat{f})$$

Example.

- $\hat{I}(f) = I(\hat{f}) = f(0) = \delta(f)$. Thus, $\hat{I} = \delta$.
- $\hat{\delta}(f) = \delta(\hat{f}) = \hat{f}(0) = I(f)$. Thus, $\hat{\delta} = I$.

The following lemma explains this apparent duality.

Lemma 8. For every $T \in \mathcal{S}'$ and $f \in \mathcal{S}$, we have $\hat{\hat{T}}(f) = T(\tilde{f})$.

Proof. $\hat{\hat{T}}(f) = \hat{T}(\hat{f}) = T(\hat{\hat{f}}) = T(\tilde{f})$. □

Recall that $\mathcal{S}$ is invariant under multiplication by functions of moderate growth. This motivates the following definition.

Definition 13. Let f be of moderate growth, and $T \in \mathcal{S}$. We define their product as

$$fT(g) := T(fg)$$

This allows us to generalize two important properties of the Fourier transform to tempered distributions.

Theorem 12. Let $T \in \mathcal{S}'$

(a) The Fourier transform takes multiplication by $-2\pi ix$ to derivative.

(b) The Fourier transform takes derivative to multiplication by $2\pi iy$.

Proof. Let $S(x) = -2\pi i x T(x)$. Then for $f \in \mathcal{S}$,

$$(\hat{T})'(f) = -\hat{T}(f') = -T(\widehat{f'}) = -T(2\pi i y \hat{f}(y)) = S(\hat{f}) = \hat{S}(f)$$

For the second part, compute

$$\widehat{T'}(f) = T'(\hat{f}) = -T(\hat{f}') = T(\widehat{2\pi i x f(x)}) = \hat{T}(2\pi i x f(x)) = 2\pi i y \hat{T}(f)$$

□

6 Topological Groups

Topological groups are the natural generalisation of the domains $(\mathbb{R}, \mathbb{Q}, \mathbb{Z}, \mathbb{T})$ on which we have been studying Fourier analysis.

Definition 14 (Topological Group)**.** A topological group is a group G equipped with a topology such that the group operation and inversion maps are continuous.

Example. Every group is a topological group under the discrete topology.

Example. $(\mathbb{R}, +), (\mathbb{R}^\times, *)$ under the usual metric and $(\mathbb{T}, *)$ under $\|\cdot\|_2$.

Our concern is with a special type of topological groups, namely the Locally Compact Abelian Groups.

6.1 LCA Groups

Let X be a topological space. A sequence of compact sets $K_n \subset K_{n+1}$ such that $X = \bigcup_n K_n$ is called a *compact exhaustion* of X.

X is called *σ-compact* if it has a compact exhaustion. Moreover, X is called *locally compact* if every point has a compact neighbourhood. Combining the definitions, we call X, *σ-locally compact* or σ-LC if it has both these properties.

Definition 15 (LCA Groups)**.** A σ-LC, abelian group is an LCA Group.

Example. $\mathbb{R}, \mathbb{T}$ and any countable abelian group with the discrete topology.

Lemma 9. Every LCA Group is separable.

Proof. Let (K_n) be a compact exhaustion. Using the Generalized Heine-Borel Theorem, cover K_n by finitely many balls of radius $\frac{1}{n}$. The centers of these balls is a countable dense subset. □

A compact exhaustion (K_n) is called *absorbing* if every compact set K eventually lies in some K_n.

Lemma 10. Every LCA group has an absorbing exhaustion.

Proof. Let (L_n) be a compact exhaustion of A and U be an open neighbourhood such that $\overline{U}$ is compact. Let $K_n = \overline{U}L_n$. Then (K_n) is absorbing.

Let $K \subset A$ be compact. Assume that for each $n \in \mathbb{N}$, there is $x_n \in K$ such that $x_n \notin K_n$. By compactness of K, choose a convergent subsequence such that $x_{n_t} \longrightarrow x \in K$. But x eventually lies in some L_n and x_n eventually lies in Ux. Hence, eventually, $x_n \in Ux \subset UL_n \subset \overline{U}L_n = K_n$. Contradiction. □

6.2 The Character Group

We define a *character* of an LCA group A is a continuous group homomorphism, $\chi : A \to \mathbb{T}$. We denote the set of characters of A by $\hat{A}$.

Definition 16 (Dual Group)**.** $\hat{A}$ forms an abelian group under pointwise product, called the *dual group*.

Proof. Routine verification. □

Example. The following are easy to verify.

(a) The characters of $\mathbb{Z}$ are given by $k \mapsto z^k$ for $z \in \mathbb{T}$ and we have $\hat{\mathbb{Z}} \cong \mathbb{T}$.

(b) The characters of $\mathbb{T}$ are given by $z \mapsto z^k$ for $k \in \mathbb{Z}$ and we have $\hat{\mathbb{T}} \cong \mathbb{Z}$.

(c) The characters of $\mathbb{R}$ are given by $x \mapsto e^{2\pi i x y}$ for $y \in \mathbb{R}$ and we have $\hat{\mathbb{R}} \cong \mathbb{R}$.

Our goal is to show that the dual group is also an LCA group. To this end, we first define a suitable metric on $\hat{A}$. Fix an absorbing compact exhaustion $A = \bigcup_n K_n$ and for $\alpha, \beta \in \hat{A}$, let

$$\hat{d}_n(\alpha, \beta) = \sup_{x \in K_n} |\alpha(x) - \beta(x)| \quad \text{and} \quad \hat{d}(\alpha, \beta) = \sum_{n=1}^{\infty} \frac{1}{2^n} \hat{d}_n(\alpha, \beta)$$

It is easy to verify that $\hat{d}$ is indeed a metric on $\hat{A}$.

Theorem 13. $(\hat{A}, \hat{d})$ is an LCA group.

Proof. We show that group operations are continuous. Let $\chi_j \to \chi$ and $\eta_j \to \eta$ in the metric. Then, for each n,

$$\sup_{x \in K_n} |\chi_j(x)\eta_j(x) - \chi(x)\eta(x)| \leq \sup_{x \in K_n} |\chi_j(x) - \chi(x)| + \sup_{x \in K_n} |\eta_j(x) - \eta(x)|$$

since all characters take values in the unit circle $\mathbb{T} \subset \mathbb{C}$. This implies that $\chi_j \eta_j \to \chi\eta$, so multiplication is continuous. Similarly, the inverse $\chi \mapsto \overline{\chi}$ is continuous, as

$$|\overline{\chi_j(x)} - \overline{\chi(x)}| = |\chi_j(x) - \chi(x)|$$

Thus, $\hat{A}$ is a topological group. It remains to show that the topology coincides with the topology of local uniform convergence, and that $\hat{A}$ is locally compact. Let $\chi_j \to \chi$ in the metric $\hat{d}$. Then, for every compact $K \subset A$, there exists n such that $K \subset K_n$. Then,

$$\sup_{x \in K} |\chi_j(x) - \chi(x)| \leq d_n(\chi_j, \chi) \to 0,$$

shows locally uniform convergence. Conversely, local uniform convergence implies $d_n(\chi_j, \chi) \to 0$ for each n, so $d(\chi_j, \chi) \to 0$.

To show local compactness, fix $\chi \in \hat{A}$ and for $\varepsilon > 0$, consider

$$B_\varepsilon(\chi) = \{\eta \in \hat{A} \mid d(\chi, \eta) < \varepsilon\}.$$

This is open by definition. One can construct a compact neighborhood of χ using equicontinuity and Arzela-Ascoli's theorem (proof omitted in book), which completes the proof. □

6.3 Pontryagin Duality

We show that every LCA group is canonically isomorphic to its bidual.

Proposition 7. If A is compact then $\hat{A}$ is discrete and vice-versa.

Proof. **Case 1: A is compact.**

We choose the exhaustion $K_1 = K_2 = \cdots = A$ and let the metric on $\hat{A}$ be sup. It suffices to show that if $d(\chi, \eta) < \sqrt{2}$, then $\chi = \eta$.

Set $\alpha = \chi^{-1}\eta$. Then $d(\alpha, 1) < \sqrt{2}$ which implies

$$|\alpha(x) - 1| < \sqrt{2} \quad \text{for all } x \in A.$$

But this means that:

$$\Re(\alpha(x)) > 0 \quad \text{for all } x \in A,$$

so the image $\alpha(A) \subset \mathbb{T}$ lies in the set $\{z \in \mathbb{T} \mid \Re(z) > 0\}$, which is a proper arc of the unit circle. But only the trivial subgroup lies in a proper arc. Thus, $\alpha = 1$ and hence, $\chi = \eta$.

Case 2: A is discrete.

Then A is countable and σ-compact. Enumerate $A = \{a_k\}_{k\in\mathbb{N}}$. Let (χ_j) be a sequence in $\widehat{A}$. For each k, the sequence $(\chi_j(a_k))$ takes values in $\mathbb{T}$, which is compact. Thus by a diagonal argument, we can extract a subsequence (χ_{j_n}) such that for every $k \in \mathbb{N}$, the sequence $(\chi_{j_n}(a_k))$ converges in $\mathbb{T}$.

Define $\chi : A \to \mathbb{T}$ by:

$$\chi(a_k) = \lim_{n\to\infty} \chi_{j_n}(a_k).$$

Then χ is a well-defined map from A to $\mathbb{T}$, and it is easy to check that it is a character (since characters are homomorphisms and the limit of homomorphisms is a homomorphism pointwise). Moreover, this pointwise convergence implies local uniform convergence on compact subsets (which are finite in the discrete topology). Therefore, $\chi_{j_n} \to \chi$ in the topology of $\widehat{A}$, showing that $\widehat{A}$ is compact. □

Theorem 14 (Pontryagin Duality)**.** Let A be an LCA group. Then

$$\alpha : A \longrightarrow \hat{\hat{A}}, \quad \alpha(a)(\chi) = \chi(a)$$

is a canonical isomorphism of topological groups.

We do not prove this, in its full generality, but we establish the slightly easier fintie case in the next section.

6.4 Finite Abelian Groups

We shall consider the special case when the group under consideration is finite. Note that all group homomorphisms, $\chi : A \to \mathbb{T}$ are continous, and hence characters. Moreover, the set of characters is also an abelian group under pointwise product. When A is cyclic, we have the following lemma.

Lemma 11. Let A be cyclic of order N. Then, $A \cong \hat{A}$.

Proof. Let $\eta \in \hat{A}$ and τ be a generator of A. The $\eta(\tau)$ satisfies $z^{o(A)} = 1$. Therefore $\eta(\tau) = e^{2\pi i \frac{l}{N}}$ for some $l \in \{0, 1, \ldots, o(A)-1\}$. The lemma follows. □

Lemma 12. Let A be a finite abelian group and $a \in A$. Suppose $\chi(a) = 1$ for $\chi \in \hat{A}$. Then $a = 1$.

Proof. By **Lemma 11**, the claim holds for cyclic groups. Assume it holds for A, B. Then for $\chi \in \widehat{A \times B}$, if $\chi(a_0, b_0) = 1$, the restriction $\chi_A(a_0) = \chi(a_0, b_0) = 1$ is a character of A, and implies $a_0 = 1$. Similarly $b_0 = 1$ and the claim holds for $\widehat{A \times B}$.

By the Fundamental Theorem of Finite Abelian Groups, the lemma follows. □

Theorem 15 (Pontryagin Duality)**.** For a finite abelian group A, there is a canonical isomorphism to the bidual $A \to \hat{\hat{A}}$ given by $a \mapsto (\chi \mapsto \chi(a))$

Proof. The map $a \mapsto \delta_a$ where δ_a is point-evaluation at a is a homomorphism, since

$$\delta_{ab}(\chi) = \chi(ab) = \chi(a)\chi(b) = \delta_a(\chi)\delta_b(\chi)$$

By **Lemma 12**, the map is injective, and since $|A| = |\hat{A}| = |\hat{\hat{A}}|$, it is also bijective. □

6.5 The Fourier Transform

Recall that for finite abelian groups, $\hat{A} \subset l^2(A)$. Using the inner product on the Hilbert space $l^2(A)$, we get the following lemma.

Lemma 13. For $\alpha, \beta \in \hat{A}$,

$$\langle \alpha, \beta \rangle = \begin{cases} |A| & \text{if} \quad \alpha = \beta \\ 0 & \text{otherwise} \end{cases}$$

Proof. The first case is easy to verify. For the second case, let $\gamma = \alpha\beta^{-1}$ and

$$\langle \alpha, \beta \rangle = \sum_{a \in A} \alpha(a)\beta(a)^{-1} = \sum_{a \in A} \gamma(a)$$

Choose $b \in A$ such that $\alpha(b) \neq 1$. Then

$$\langle \alpha, \beta \rangle \gamma(b) = \sum_{a \in A} \gamma(a)\gamma(b) = \sum_{a \in A} \gamma(ab) = \sum_{a \in A} \gamma(a) = \langle \alpha, \beta \rangle$$

which yields $\langle \alpha, \beta \rangle = 0$. □

One can now define Fourier transform as a member of the bidual of A. For $f \in l^2(A)$, we define its *Fourier transform* $\hat{f} : \hat{A} \to \mathbb{C}$ by

$$\hat{f}(\chi) = \frac{1}{\sqrt{|A|}} \langle f, \chi \rangle$$

The factor of $\frac{1}{\sqrt{|A|}}$ normalizes the inner products of $l^2(A)$ and $l^2(\hat{A})$ such that the Fourier transform turns out to be an isometry. In fact, it is even stronger as we show next.

Theorem 16. The map $f \mapsto \hat{f}$ is an isomorphism of the Hilbert spaces $l^2(A) \to l^2(\hat{A})$.

Proof. Let $f, g \in l^2(G)$. Using **Lemma 12**,

$$\begin{aligned}
\langle \hat{f}, \hat{g} \rangle &= \sum_{\chi \in \hat{G}} \hat{f}(\chi)\overline{\hat{g}(\chi)} \\
&= \sum_{\chi \in \hat{G}} \left(\frac{1}{\sqrt{|G|}} \sum_{a \in G} f(a)\overline{\chi(a)} \right) \overline{\left(\frac{1}{\sqrt{|G|}} \sum_{b \in G} g(b)\overline{\chi(b)} \right)} \\
&= \frac{1}{|G|} \sum_{a,b \in G} f(a)\overline{g(b)} \sum_{\chi \in \hat{G}} \overline{\chi(a)}\chi(b) \\
&= \frac{1}{|G|} \sum_{a,b \in G} f(a)\overline{g(b)} \langle \delta_b, \delta_a \rangle \\
&= \sum_{a \in G} f(a)\overline{g(a)} = \langle f, g \rangle
\end{aligned}$$

Thus, the Fourier Transform is an isometry, hence injective. Since, the Hilbert spaces are of equal dimension, the map is an isomorphism. □

Corollary 6. $\hat{\hat{f}}(\delta_a) = f(a^{-1})$

Proof. By direct computation,

$$\begin{aligned}
\hat{\hat{f}}(\delta_a) &= \frac{1}{\sqrt{|G|}} \sum_{\chi \in \hat{G}} \hat{f}(\chi)\overline{\delta_a(\chi)} \\
&= \frac{1}{\sqrt{|G|}} \sum_{\chi \in \hat{G}} \left(\frac{1}{\sqrt{|G|}} \sum_{b \in G} f(b)\overline{\chi(b)} \right) \overline{\chi(a)} \\
&= \frac{1}{|G|} \sum_{\chi \in \hat{G}} \sum_{b \in G} f(b)\overline{\chi(b)\chi(a)} \\
&= \frac{1}{|G|} \sum_{\chi \in \hat{G}} \sum_{b \in G} f(b^{-1})\chi(b)\overline{\chi(a)} \\
&= \frac{1}{|G|} \sum_{b \in G} f(b^{-1})\langle \delta_b, \delta_a \rangle \\
&= f(a^{-1})
\end{aligned}$$

□

6.6 Convolution

One can naturally define an analog of convolution product on $L^1_{bc}(\mathbb{R})$ for $l^2(A)$. Let $f, g \in l^2(A)$. We define their *convolution product* by

$$f * g(a) = \frac{1}{\sqrt{|A|}} \sum_{b \in A} f(b)g(b^{-1}a)$$

In fact, the two above definitions for Fourier transform and convolutions, behave very similarly to the case for $\mathcal{S}$ as is evident from the next theorem. It is easy to show that for $f, g, h \in l^2(A)$, we have $\langle f * g, h \rangle = \frac{1}{\sqrt{|A|}} \langle f, h \rangle \langle g, h \rangle$.

Theorem 17. The Fourier transform takes convolutions to pointwise product.

Proof. By direct computation,

$$\widehat{f * g}(\chi) = \frac{1}{\sqrt{|A|}} \langle f * g, \chi \rangle = \frac{1}{|A|} \langle f, \chi \rangle \langle g, \chi \rangle = \hat{f}(\chi)\hat{g}(\chi)$$

□

Property. The following are immediate consequences of the above theorem.

$$f * g = g * f \qquad f * (g * h) = (f * g) * h \qquad f * (g + h) = f * g + f * h$$

7 Plancherel Theorem

In order to prove Plancherel's Theorem, we generalize the notion of Riemann Integral, to the Haar Integral for groups. However this definition is imprecise and is only made as per the requirements. A formal and precise defintion needs measure-theoritic tools.

7.1 Haar Integration

We generalize the notion of Riemann Integrals to groups, specifically to LCA groups. Let $C_c(G)$ be the *space* of all continuous functions $f : G \to \mathbb{C}$, with compact support. We will define the inner product later. A linear functional $I : C_c(G) \to \mathbb{C}$ is called an *integral* if for $f \in C_c(G)$

$$f \geq 0 \quad \Rightarrow \quad I(f) \geq 0$$

Note that if the choice of I is clear from the context, we will simply write $I(f) = \int_G f(x)dx$. We state the following theorem without proof.

Theorem 18. Let G be an arbitrary LCA group. Then there exists a non-zero invariant integral I of G, unique upto scaling.

Any such invariant integral is called a Haar integral. Note that $C_c(G)$ forms a *space* with the inner product,

$$\langle f, g \rangle = \int_G f(x)\overline{g(x)}dx$$

and the Hilbert space completion of $C_c(G)$ is denoted as $L^2(G)$. We give some examples of Haar integrals.

Example. Haar integrals.

- On $\mathbb{R}$,
$$I(f) = \int_{-\infty}^{\infty} f(x)dx$$
- On $\mathbb{T}$,
$$I(f) = \int_0^1 f(x)dx$$
- On $\mathbb{R}_+^\times$,
$$I(f) = \int_0^\infty f(x)\frac{dx}{x}$$

These examples show that the integrals used in the definition of the Fourier Series and Fourier Transform were actually Haar integrals on the correspoding LCA groups.

7.2 General Plancherel Theorem

The following lemma will be needed later.

Lemma 14. Fix a Haar integral for the compact abelian group A, such that $\int_A 1dx = 1$. Then for $\chi, \eta \in \hat{A}$, we have

$$\int_A \chi(x)\overline{\eta(x)} = \begin{cases} 1 & \text{if } \chi = \eta \\ 0 & \text{otherwise} \end{cases}$$

Proof. The proof is similar to the proof of **Lemma 13**. □

Theorem 19. Let A be an LCA group. There is a unique Haar measure on $\hat{A}$ such that for every $f \in \frac{1}{bc}(A)$,

$$\|f\|_2 = \|\hat{f}\|_2$$

The Fourier transform extends to a Hilbert space isomorphism of the completions $L^2(A) \to L^2(\hat{A})$.

Specializing this above theorem to the unit torus, $\mathbb{T}$, we get that for $f \in C(\mathbb{T})$,

$$\int_0^1 |f(x)|^2 dx = \|f\|_2^2 = \|\hat{f}\|_2^2 = \sum_{k\in\mathbb{Z}} \hat{f}(k)^2 = \sum_{k\in\mathbb{Z}} |c_k(f)|^2$$

we get the Parseval Identity.

Proof. We only prove this for the special case of a discrete group. Let $f \in L^2(A)$. Define the function $\tilde{f}(x) := \overline{f(-x)}$ on A. Since A is abelian and discrete, the Haar integral is simply a sum over A. Define $g := \tilde{f} * f$ that is,

$$g(x) = \sum_{y\in A} \tilde{f}(y)f(x-y) = \sum_{y\in A} \overline{f(-y)}f(x-y).$$

Computing the Fourier transform of g:

$$\widehat{g}(\chi) = \widehat{\tilde{f} * f}(\chi) = \widehat{\tilde{f}}(\chi) \cdot \widehat{f}(\chi)$$

by the convolution theorem. Computing $\widehat{\tilde{f}}(\chi)$:

$$\widehat{\tilde{f}}(\chi) = \sum_{x\in A} \overline{f(-x)\chi(x)} = \sum_{x\in A} \overline{f(-x)\chi(x)}$$

Making the substitution $y = -x$,

$$\widehat{\tilde{f}}(\chi) = \sum_{y\in A} \overline{f(y)\chi(-y)} = \sum_{y\in A} \overline{f(y)}\chi(y) = \overline{\widehat{f}(\chi)}.$$

Therefore, $\widehat{g}(\chi) = \overline{\widehat{f}(\chi)} \cdot \widehat{f}(\chi) = |\widehat{f}(\chi)|^2$. Now, using the Fourier inversion formula, we get

$$g(0) = \int_{\widehat{A}} \widehat{g}(\chi)\, d\chi = \int_{\widehat{A}} |\widehat{f}(\chi)|^2\, d\chi$$

On the other hand,

$$g(0) = \sum_{y \in A} \overline{f(-y)} f(-y) = \sum_{y \in A} |f(-y)|^2 = \sum_{x \in A} |f(x)|^2$$

since $x = -y$ just reindexes the sum. Therefore,

$$\sum_{x \in A} |f(x)|^2 = \int_{\widehat{A}} |\widehat{f}(\chi)|^2\, d\chi.$$

This proves the Plancherel identity in the case where A is discrete. □

This truly is a generalization of the Plancherel Theorem that we saw in the case of Fourier Transform on $\mathbb{R}$.

8 References

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